

Math 300

Quiz # 4

Key

Use the given Cayley table to answer the following questions 1, 2, and 3.

*	0	1	2	3	4	5	6	7	8	9
0	0	1	2	3	4	5	6	7	8	9
1	1	2	3	4	0	6	7	8	9	5
2	2	3	4	0	1	7	8	9	5	6
3	3	4	0	1	2	8	9	5	6	7
4	4	0	1	2	3	9	5	6	7	8
5	5	9	8	7	6	0	4	3	2	1
6	6	5	9	8	7	1	0	4	3	2
7	7	6	5	9	8	2	1	0	4	3
8	8	7	6	5	9	3	2	1	0	4
9	9	8	7	6	5	4	3	2	1	0

1. a) $8*6 = 2$

b) $9*8 = 1$

2. a) What is the identity element in this group?

0

b) What is the inverse of 4?

1

3. List all elements in the centralizer of 2.

$$C(2) = \{0, 1, 2, 3, 4\}$$

4. List all the elements in the group $U(9)$.

$$U(9) = \{1, 2, 4, 5, 7, 8\}$$

5. How many elements are there in each of the following groups?

$$|Z_{10}| = 10$$

$$|U(10)| = 4$$

$$|S_{10}| = 10!$$

$$|D_{10}| = 20$$

$$\{1, 3, 7, 9\}$$

$$3, 628, 800$$

Over →

6. How do you compute the order of a permutation written in standard cycle form?

$|\alpha|$ = least common multiple of the lengths
of the disjoint cycles

7. Let $\alpha = (1\ 2\ 3\ 4\ 5\ 6\ 7\ 8)$

a) Write α^4 in standard cycle form.

$$\alpha^4 = (1\ 5)(2\ 6)(3\ 7)(4\ 8)$$

b) Write α^{-1} in standard cycle form.

$$\alpha^{-1} = (1\ 8\ 7\ 6\ 5\ 4\ 3\ 2)$$

8. Write the permutation $(1\ 3\ 5\ 2\ 4)(6\ 4\ 2\ 3)(1\ 5\ 3)(2\ 5\ 4)$ in standard form.

$$(1\ 2\ 6)(4\ 5)$$

9. Write an if-then statement that defines what it means for a group G to be closed under its operations.

If x and y are in G then xoy is in G .